

Spreadsheet Data

- 60 images
- 30 students
- 10 images per student
- 5 students evaluated each image
- 300 total human decisions
- 30 AI-generated images
- 30 real images
- One confidence value is missing, so confidence analysis uses **299 responses**
- The two blank separator rows in Sheet 1 were ignored

1. Overall Human Performance

Using **every individual student response**:

Metric	Result
Accuracy	52.67%
Balanced Accuracy	52.67%
Precision (AI)	52.33%
Recall / Sensitivity	60.00%
Specificity	45.33%
F1-Score	55.90%

Interpretation

Overall, students correctly classified **158 out of 300 decisions**.

This suggests that the participants performed **only slightly above chance level**, meaning distinguishing AI-generated images from real images was difficult.

2. Confusion Matrix

Considering **AI** as the **positive class**:

Actual \ Predicted	AI	Real
Actual AI	90	60
Actual Real	82	68

So:

- **TP = 90** → AI image correctly identified as AI
- **FN = 60** → AI image incorrectly identified as Real
- **FP = 82** → Real image incorrectly identified as AI
- **TN = 68** → Real image correctly identified as Real

Important observation

Students were **better at identifying AI images** than identifying real images:

- AI Recall = **60%**
- Real-image Specificity = **45.33%**

This indicates a tendency to label images as **AI**, even when they were actually real.

3. Majority-Vote Accuracy

For each image, 5 students gave an answer. I used the majority decision.

Result	Value
Total images	60
Images correctly classified by majority	31
Majority-vote accuracy	51.67%

Interestingly, majority voting did **not substantially improve performance**.

This means that students often made **similar mistakes on the same images**. Their errors were not completely random.

4. Confidence Analysis

Metric	Result
Average confidence	4.13 / 5
Confidence when correct	4.17 / 5
Confidence when incorrect	4.09 / 5

This is a very interesting result

Students were highly confident **whether they were right or wrong**.

The difference is very small:

- Correct answers: **4.17**
- Incorrect answers: **4.09**

So, higher confidence did **not strongly correspond to higher accuracy**.

The confidence-correctness correlation is approximately **0.048**, which is extremely weak.

Strong evidence of overconfidence

There were **130 responses with maximum confidence (5/5)**.

Of those:

- Correct: **70**
- Incorrect: **60**
- Accuracy: **53.85%**

So even when students were **completely confident**, they were only correct about **54% of the time**.

This could become one of the strongest findings in your paper.

5. Accuracy by Confidence Level

Confidence	Responses	Accuracy
1	1	100.00%*
2	9	33.33%
3	70	48.57%
4	89	56.18%
5	130	53.85%

*Only one response, so it should not be interpreted heavily.

The key finding is that **confidence level 5 was not the most accurate group**.

6. Per-Category Accuracy

Category	Individual Accuracy	Majority-Vote Accuracy
Aeroplane	56.67%	50.00%
Car	46.67%	50.00%
Bird	26.67%	33.33%
Cat	50.00%	50.00%
Deer	56.67%	50.00%
Dog	80.00%	83.33%
Frog	43.33%	33.33%
Horse	63.33%	50.00%
Ship	43.33%	50.00%
Truck	60.00%	66.67%

Best category: 🐕 Dog

- Individual accuracy: **80%**
- Majority-vote accuracy: **83.33%**

Worst category: Bird

- Individual accuracy: **26.67%**
- Majority-vote accuracy: **33.33%**

This means the bird images were particularly difficult for participants to distinguish.

7. Per-Image Difficulty

The images showed a large variation in difficulty:

- **11 images** were correctly identified by all 5 students → **100% accuracy**
- **9 images** were incorrectly identified by all 5 students → **0% accuracy**
- **19 images** had only 0–20% individual accuracy
- **24 images** had 80–100% individual accuracy

This is important because it shows that **human ability was not consistent across images**. Some AI and real images were extremely easy to identify, while others consistently fooled participants.

8. Individual Student Performance

The best-performing students achieved:

- **80% accuracy**

The lowest-performing student achieved:

- **20% accuracy**

This shows substantial variation between participants.

The Main Result for Your Paper

A strong summary would be:

The study recorded 300 individual human classification decisions across 60 images, evenly distributed between AI-generated and real images. Participants achieved an overall accuracy of 52.67%, with a balanced accuracy of 52.67%. While the recall for AI-generated images was 60.00%, the specificity for real images was lower at 45.33%, indicating a tendency among participants to classify images as AI-generated. Participants reported high confidence in their decisions (mean = 4.13/5); however, confidence showed a very weak relationship with correctness. Majority voting across five participants per image resulted in an accuracy of 51.67%, suggesting that participant errors were often shared rather than random.

One important point

You have **two different valid accuracy results**:

- **52.67% = individual human decision accuracy** → best for the main overall human-performance analysis.
- **51.67% = majority-vote image accuracy** → useful for analyzing how groups performed on individual images.

I recommend reporting both, because they answer slightly different research questions.

I can next turn this spreadsheet analysis into a **complete “Results and Discussion” section for your research paper**, including properly formatted tables, interpretations, and graphs.